

Harsh Karekar

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Professional Summary

Final-year ECE undergraduate with a strong foundation in AI, Machine Learning, and Data Science. Skilled in designing intelligent systems, analyzing complex datasets, and building scalable, production-ready solutions. Experienced in translating real-world problems into deployed applications. Seeking opportunities as an AI Engineer or ML Engineer.

Education

Vellore Institute of Technology | Bhopal 2022 – 2026 (Expected)
B.Tech in Electronics and Communication Engineering | CGPA: 8.39 / 10

Technical Skills

Languages: Python, Java, SQL
ML & Deep Learning: XGBoost, Random Forest, Autoencoder (TensorFlow/Keras), Scikit-learn, NLP, RAG Workflows, Agentic AI, LLMs
Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Jupyter
MLOps & Deployment: FastAPI, REST API Design, Model Serving, Cloud Deployment (Render, Vercel), Git/GitHub
Web & Tools: React.js, Node.js, Tailwind CSS

Projects

- FraudShield** | [Live Demo](#) | Python, XGBoost, Autoencoders, TensorFlow/Keras, FastAPI, React Aug – Nov 2025

 - Engineered a hybrid fraud detection system combining supervised XGBoost and an unsupervised Deep Autoencoder to detect known fraud signatures and zero-day anomalies across 284,807 transactions with only 0.17% fraud rate — a highly imbalanced real-world setting.
 - Designed a Hybrid Risk Score ($\alpha \times \text{XGBoost prob} + (1-\alpha) \times \text{anomaly score}$, scaled 0–100%) with fraud flagging at >50% threshold, achieving strong generalization on unseen fraud patterns.
 - Built and deployed a full-stack ML inference pipeline — FastAPI REST backend (Render) with batch inference and a React + TypeScript dashboard (Vercel) — serving real-time and batch fraud analysis.
 - Handled extreme class imbalance by training the anomaly model exclusively on legitimate transactions, improving detection of evolving fraud not seen during supervised training.
- PATLens – Placement Manager** | Python · LLM (Mistral) · Gmail API · Google Sheets API Feb – May 2025

 - Built an end-to-end agentic AI pipeline that autonomously ingests unstructured placement emails from Gmail, extracts 15 structured fields (company, CTC, eligibility, deadlines, links) via LLM-based semantic parsing, and maintains a live Google Sheets tracker — zero manual effort.
 - Replaced brittle regex extraction with LLM + deterministic post-processing, enabling reliable parsing of inconsistent human-written email formats with modular, cron-schedulable architecture.
 - Designed an incremental append-only data pipeline with scheduled execution (cron/Task Scheduler), supporting scalable extension for eligibility checks and analytics.
- Trader Performance vs Bitcoin Market Sentiment** | Python, Pandas, Scikit-learn, Streamlit Dec 2025 – Feb 2026

 - Analyzed 10,000+ real trade executions from Hyperliquid DEX against the Bitcoin Fear & Greed Index, uncovering that traders are measurably more profitable during Fear regimes than Greed periods.
 - Trained a Random Forest classifier on sentiment score, trade size, and direction (BUY/SELL) to predict trade-level profitability, demonstrating predictability using market sentiment as a feature.
 - Built and deployed an interactive Streamlit dashboard with regime-based PnL charts, wallet leaderboard, BUY vs SELL profitability breakdown, and an AI-powered trade profitability predictor.

Certifications

- IBM Data Science Professional Certificate | IBM / Coursera 2024
- OCI Certified Generative AI Professional | Oracle 2025
- Generative AI using Watsonx | IBM 2024
- Artificial Intelligence Fundamentals | IBM 2024

Extracurricular

- Technical Team Representative – Linux Club, VIT Bhopal** Feb – Apr 2024

 - Represented the technical team in organizing a competitive CTF event focused on Linux fundamentals, terminal navigation, and cybersecurity challenges, managing event infrastructure for 150+ participants.
- Event Management Lead – Matrix, The Multimedia Club, VIT Bhopal** Jan – Jun 2025

 - Led planning and execution of 2 major college-level multimedia events, managing 500+ participants and coordinating with 40+ cross-functional team members to ensure smooth operations.